Language JEPA for Reasoning: Predicting Answer Embeddings from Questions and Paraphrase Pairs

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Abstract—Autoregressive language models are trained to predict tokens, which couples the objective to surface form. We study a language joint-embedding predictive architecture that predicts the embedding of an answer from the embedding of a question, and that uses paraphrase pairs as an additional signal. We compare it with an autoregressive objective on arithmetic and multi-step reasoning tasks, and on a synthetic arithmetic setting where the correct answer is known by construction. A symbolic pilot shows that a small MLP readout fails beyond depth 2; the language JEPA results are pending.

Index Terms—language models, JEPA, reasoning, paraphrase, embedding prediction

I. Introduction

Reasoning tasks reward the same answer under many surface forms. An objective that predicts answer embeddings rather than tokens may encourage representations invariant to paraphrase, but it may also lose the precision needed to produce an exact answer. We ask whether a language JEPA improves multi-step reasoning under matched compute, and whether paraphrase supervision is the cause of any gain.

II. Related Work

The author's recent artificial intelligence publications on self-training, clinical deep learning, curriculum learning, model merging, synthetic data, long-context training, preference alignment, direct preference optimization, reinforcement learning with verifiable rewards, distillation, mixture-of-experts, and parameter-efficient fine-tuning [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] form the background for this work.

JEPA-style prediction in embedding space is introduced for images and video [13]. Its use for language is less studied. The author's recent work on Mixture-of-Experts training [9], long-context training [6], and parameter-efficient fine-tuning [10] provides the training setting used here. Preference-based alignment methods [7] are relevant as an alternative route to behavior control.

III. Method

Given a question q and an answer a, an encoder f_θ produces $z_q = f_\theta(q)$ and a target encoder $\bar{f}$ produces $z_a = \bar{f}(a)$. The predictor g_ϕ is trained so that $g_\phi(z_q) \approx z_a$. Paraphrase pairs (q, q') add a term that asks $g_\phi(f_\theta(q))$ and $g_\phi(f_\theta(q'))$ to agree. Decoding for an exact answer uses a small readout head trained on frozen predictions.

Equations and their meaning

The synthetic task has operators $o_i \in \{-1, +1\}$ and values $v_i \in \{0, \ldots, 9\}$ for $i = 1, \ldots, k$. The answer is

$$y = \Big(\sum_{i=1}^{k} o_i \, v_i \Big) \bmod 10, \tag{1}$$

so reasoning depth k is the number of steps that must be combined.

The autoregressive baseline minimizes the negative log-likelihood of the answer tokens a_t given the question q and previous tokens:

$$\mathcal{L}_{\mathrm{AR}} = -\sum_{t} \log p_\theta\big(a_t \mid q, a_{<t}\big). \tag{2}$$

The language JEPA predicts the answer embedding from the question embedding:

$$\mathcal{L}_{\mathrm{JEPA}} = \big\| g_\phi(f_\theta(q)) - \bar{f}(a) \big\|_2^2, \tag{3}$$

where f_θ encodes the question, $\bar{f}$ is the stop-gradient target encoder for the answer, and g_ϕ is the predictor. Paraphrase pairs (q, q') add an agreement term $\|g_\phi(f_\theta(q)) - g_\phi(f_\theta(q'))\|_2^2$, which asks paraphrases to map to the same prediction.

The readout accuracy over N test items is

$$\mathrm{Acc} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\big[\hat{y}_n = y_n\big], \tag{4}$$

where $\mathbf{1}[\cdot]$ is 1 when the predicted answer $\hat{y}_n$ equals the true answer y_n and 0 otherwise. Chance level for ten classes is 0.1.

IV. Experimental Protocol

Synthetic arithmetic. Multi-step integer problems with a known solution, with depth $k \in \{1, 2, 3, 4\}$, to measure accuracy as a function of reasoning depth.

Benchmarks. GSM8K-style word problems in English and Portuguese translations, with exact-match accuracy.

Baselines. Autoregressive fine-tuning with the same backbone and token budget; the same model without paraphrase pairs.

Compute. Matched number of training tokens and parameters. Compute is reported in FLOPs.

Statistics. Five seeds, 95% bootstrap intervals, paired tests.

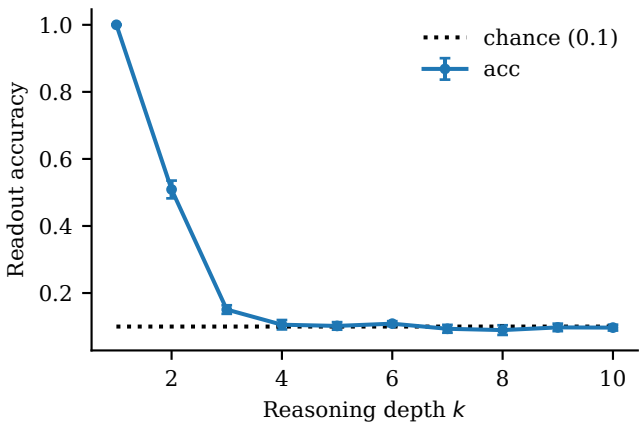


Fig. 1. Readout accuracy versus reasoning depth on synthetic arithmetic (pilot, MLP). Dotted line: chance level.

V. Results

This pilot arm uses symbolic arithmetic with a small MLP readout, not a language JEPA. Accuracy is 1.00 at depth $k = 1$, 0.51 at $k = 2$, and at chance level (about 0.10) for $k \geq 3$. The readout fails beyond shallow depth, so this arm gives no evidence about reasoning. The autoregressive baseline, the paraphrase ablation, and the language benchmarks remain pending.

VI. Discussion and Limitations

The readout head adds a decoding step that could account for part of the difference, so the protocol reports results with and without it. Translation may change difficulty, so Portuguese and English are reported separately.

VII. Conclusion

We propose a matched-compute comparison of answer-embedding prediction and autoregressive training on reasoning, with paraphrase supervision as an explicit ablation.

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